

Rethinking Spectral Graph Neural Networks With Spatially Adaptive Filtering

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Abstract—Whilst spectral graph neural networks (GNNs) are theoretically well-founded in the spectral domain, their practical reliance on polynomial approximation implies a profound linkage to the spatial domain. As previous studies rarely examine spectral GNNs from the spatial perspective, their spatial-domain interpretability remains elusive, e.g., what information is essentially encoded by spectral GNNs in the spatial domain? In this article, to answer this question, we investigate the theoretical connection between spectral filtering and spatial aggregation, unveiling an intrinsic interaction that spectral filtering implicitly leads the original graph to an adapted new graph, explicitly computed for spatial aggregation. Both theoretical and empirical investigations reveal that the adapted new graph not only exhibits nonlocality but also accommodates signed edge weights to reflect label consistency among nodes. These findings highlight the interpretable role of spectral GNNs in the spatial domain and inspire us to rethink graph spectral filters beyond the fixed-order polynomials, which limit the effective propagation range and hinder their ability to capture long-range dependencies. Built upon the theoretical findings, we revisit the state-of-the-art spectral GNNs and propose a novel spatially adaptive filtering (SAF) framework, which leverages the adapted new graph by spectral filtering for an auxiliary nonlocal aggregation. Notably, our SAF comprehensively models both node similarity and dissimilarity from a global perspective, therefore alleviating persistent deficiencies of GNNs related to long-range dependencies and graph heterophily. Extensive experiments over 13 node classification benchmarks demonstrate the superiority of our proposed framework to the state-of-the-art methods.

Index Terms—Graph heterophily, graph neural networks (GNNs), long-range dependency, spectral filtering.

NOMENCLATURE

$\mathcal{G}$	Graph.
$\mathcal{V}, \mathcal{E}$	Node and edge set.
N, C	Node and Class number.
F, d	Raw and hidden feature number.

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m	Extremal eigenvalue number.
Δ	Longest geodesic node distance on the graph.
$\mathcal{H}, \mathcal{H}_{\text{class}}, \mathcal{H}_{\text{adjusted}}$	Edge, class, and adjusted homophily ratios.
$\mathbf{A}, \hat{\mathbf{A}} \in \mathbb{R}^{N \times N}$	Adjacency matrix and its normalized version.
$\mathbf{D}$	Degree matrix.
$\mathbf{L}, \hat{\mathbf{L}} \in \mathbb{R}^{N \times N}$	Laplacian matrix and its normalized version.
$\mathbf{U}, \mathbf{\Lambda} \in \mathbb{R}^{N \times N}$	Eigenvector and eigenvalue matrixes.
$\hat{\mathbf{A}}^{\text{new}} \in \mathbb{R}^{N \times N}$	Adjacency matrix of the new computation graph.
$\mathbf{X} \in \mathbb{R}^{N \times F}$	Raw node feature.
$\mathbf{Z} \in \mathbb{R}^{N \times d}$	Hidden node feature.
$\mathbf{y}_i \in \mathbb{R}^{1 \times C}$	Class prediction of node i .
$g_\phi : [0, 2 \rightarrow \mathbb{R}]$	Graph filter function.
$\gamma_\theta : [0, 2 \rightarrow \mathbb{R}]$	Function defined on graph spectrum.
$P_k(\cdot)$	k th order polynomial basis.
ϕ_k, ω_k, π_k	k th order polynomial coefficients.
$\ \cdot\ , \ \cdot\ _F^2$	Cardinality and L2 norm of a set.
$\text{tr}(\cdot)$	Trace operation.
$\text{nnz}(\cdot)$	Number of nonzero element.
L, τ	Aggregation step and scaling parameter.
η, ϵ	Update rate and threshold value.

I. INTRODUCTION

GRAPH neural networks (GNNs) have shown remarkable abilities to uncover the intricate dependencies within graph-structured data, and achieved tremendous success in graph machine learning [1], [2], [3], [4], [5], [6], [7], [8]. Spectral GNNs are a class of GNNs rooted in spectral graph theory [9], [10], implementing graph convolutions via spectral filters [11], [12]. Whilst various spectral filtering strategies [13], [14], [15], [16], [17], [18], [19], [20], [21] have been proposed for spectral GNNs, their practical implementations invariably resort to approximating graph filters with fixed-order polynomials for computational efficiency [17], [18]. This truncated approach essentially relies on the direct extraction of spatial features from the local regions of nodes. As such, the spatial domain of a graph, albeit loosely connected to spectral GNNs in theory, still plays a crucial role in effectively learning node representations.

However, there is a notable lack of research examining spectral GNNs from the spatial perspective. Though recent studies analyze both spectral and spatial GNNs to elucidate their similarities in model formulations [22], [23], [24], outcomes [25], [26], and expressiveness [18], [27], [28], [29], they neglect the interpretability that could arise mutually from the other domain. Specifically, while most spectral GNNs have well explained their learned filters in the spectral domain [15], [17], [18], [19], understandings from the spatial viewpoint are merely limited to fusing multiscale graph information [30]; this unfortunately lacks a deeper level of interpretability in the vertex domain. Therefore, a natural question arises: *what information is essentially encoded by spectral GNNs in the spatial domain?*

In this work, we attempt to answer this question by exploring the connection between spectral filtering and spatial aggregation. The former is the key component in spectral GNNs, while the latter is closely associated with spatial GNNs utilizing recursive neighborhood aggregation. In existing GNN frameworks, these two approaches rarely interact each other at the risk of domain information tradeoffs due to uncertainty principles [31], [32], [33]. Recognizing the spatial significance in spectral filtering, [15] have recently considered nonnegative constraints as part of a generalized graph optimization problem. Notably, however, spatial aggregation meanwhile resembles the optimizing trajectory of the same optimization problem through iterative steps, which may be easily overlooked. Inspired by such observation, we examine, for the first time, the theoretical interaction between spectral filtering and spatial aggregation. This exploration has led us to uncover an intriguing theoretical interplay, i.e., spectral filtering implicitly modifies the original graph, transforming it into a new one that explicitly functions as a computation graph for spatial aggregation. Delving deeper, we discover that the adapted new graph enjoys some desirable properties, enabling a direct link among nodes that originally require multiple hops to do so, thereby exhibiting nice non-locality. Moreover, we find that the new graph edges allow signed weights, which turns out capable of distinguishing between label agreement and disagreement of the connected nodes.

Overall, these findings underscore the interpretable role and significance of spectral GNNs in the spatial domain, inspiring us to rethink graph spectral filters beyond fixed-order polynomials, which, albeit efficient, limit models' effective propagation range and hinder their ability to capture long-range dependencies. Concretely, we propose a novel spatially adaptive filtering (SAF) framework to fully explore spectral GNNs in the spatial domain. SAF leverages the adapted new graph by spectral filtering for auxiliary spatial aggregation and allows individual nodes to flexibly balance between spectral and spatial features. By performing nonlocal aggregation with signed edge weights, SAF adeptly overcomes the limitations of truncated polynomials, enabling the model to capture both node similarity and dissimilarity at a global scale. As a benefit, it can mitigate persistent deficiencies of GNNs regarding long-range dependencies and graph heterophily. The contributions are summarized as follows.

- 1) Our investigation into spectral GNNs in the spatial domain reveals that they fundamentally alter the original graph, introducing nonlocality and signed edge weights to discern node label consistency.
- 2) We propose an SAF framework, a paradigm-shifting approach to spectral GNNs that jointly leverages graph learning in both spatial and spectral domains, making it a powerful tool for capturing long-range dependencies and handling graph heterophily.
- 3) Extensive experiments in node classification exhibit notable improvements of up to 15.37%, and show that SAF beats the best-performing spectral GNNs on average.

II. MOTIVATION AND RELATED WORKS

This section outlines the motivation behind our research, derived from thoughtful consideration of the existing related works, and elucidates how our work diverges from and contributes to the current body of research.

A. Graph Neural Networks

GNNs have received widespread attention from the research community and have been successfully applied across a variety of domains [34], [35], [36]. Broadly speaking, they can be divided into spatial-based and spectral-based methods. Spatial GNNs leverage the spatial connections among nodes to perform message passing, also known as spatial aggregation [37], [38] (readers are directed to works [2] and [39] for a thorough review). Spectral GNNs leverage the graph's spectral domain for convolution or, alternatively, spectral filtering [12], [40]. Prevailing approaches focus on developing polynomial graph filters, by either learning polynomial coefficients, such as GPR-GNN [14], BernNet [15], ChebNetII [17], and JacobiConv [18], or optimizing the polynomial basis for better adaptation, as seen in models like LON-GNN [20] and OptBasisGNN [21]. Diverging from this trend, ARMA [41] employs rational filter functions while still approximating them with polynomials. From a different perspective, [42] explores graph framelets to construct graph convolutions that better capture multiscale low- and high-frequency information. Building upon this, [43] further extends the framelet transform to incorporate permutation equivariance, thereby enhancing GNNs' capacity to model heterophilous graphs. Although these methods are theoretically rooted in the spectral domain, their practical reliance on polynomial approximations or reconstructing the computation graph via framelet transforms hints at a profound linkage to the spatial domain. However, the spatial-domain interpretation of spectral GNNs is rarely examined. To this end, we delve into in this article the intrinsic information spectral GNNs convey within the spatial context.

B. Unified Viewpoints for GNNs

Several works have explored the nuances between spatial and spectral GNNs. Early studies by [22] and [23] examined their similarities in model formulations. Chen et al. [27] proved their spatial GNN's anti-oversmoothing ability via

spectral analysis. Ma et al. [25] and Zhu et al. [26] utilized the graph signal denoising problem to integrate both GNN types, and their expressiveness equivalence is further explored in works [18], [28]. Recently, [29] has highlighted the feature space constraints of both spatial and spectral GNNs, while [24] tends to emphasize their relationship via residual connection. Though these studies effectively bridge spectral and spatial GNNs, they remain focused on congruencies. Unlike them, our work represents the first endeavor to delve into the interpretability of spectral GNNs in the spatial domain, emphasizing the theoretical synergy between spectral filtering and spatial aggregation. The empirical success of our method (as compared to unified GNNs in Sections VI-B and VI-C), stem from this in-depth analysis, further underscoring our practical contributions to the literature.

C. Long-Range Dependencies

While substantial efforts have been directed toward capturing long-range dependencies in spatial GNNs [27], [44], [45], [46], [47], [48], [49], the exploration of the same challenge in spectral GNNs remains understudied. Specifically, most spectral GNNs approximate graph filters with fixed-order polynomials, which, albeit efficient, limit the effective propagation range and hinder their ability to capture the long-range dependencies. To fill this gap, we propose an SAF framework that emerges as a valuable consequence of analyzing spectral GNNs in the spatial domain. Aligned with our objective, Specformer [50] is introduced to address long-range dependencies for spectral GNNs, using a transformer-based set-to-set spectral filter. However, it lacks spatial-domain interpretability and introduces more trainable parameters. In contrast, our approach creates a nonlocal new graph without learning additional parameters, meanwhile elucidating the spatial implications of spectral GNNs. Similarly, a recent approach, FLODE [51], also produces matrices capable of capturing long-range dependencies by utilizing the fractional graph Laplacian [52]. However, this work mainly generalizes existing concepts from undirected to directed graphs to mitigate graph oversmoothing. Our study, conversely, focuses on exploring the fundamental issues surrounding undirected graphs to delve deeper into the intrinsic significance of spectral GNNs.

D. Graph Heterophily

Graph heterophily [46], [53], where different labeled nodes connect, challenges GNNs operating under the homophily assumption [54]. Although many GNNs have been crafted to manage heterophilic connections [14], [43], [55], [56], [57], [58], [59], [60], [61], our method offers a distinct solution. Specifically, SAF innovatively conducts an auxiliary nonlocal aggregation using signed edge weights, emphasizing both intraclass similarity and interclass difference on a global scale. One should note that there are two recent works [62], [63] also employ signed edge weights, introducing GloGNN and LRGNN, respectively. GloGNN aims to capture global homophily but is limited to K-hop neighborhood information, while LRGNN extends this by using low-rank properties to approximate the true global relationships between node labels.

Despite their success, both methods focus primarily on fitting node label relationships without fully exploring the fundamental GNN mechanisms, leading to a heavy reliance on label supervision (see Section VI-H). Differently, our SAF leverages cross-domain insights into GNNs, ensuring the theoretical soundness of nonlocal learning, as proven in Section IV-B. This allows SAF to consistently perform well under both sparse and dense supervision. In addition, we note several recent efforts that also tackle graph heterophily from the spectral perspective. For instance, [64] proposes an ensemble spectral filtering to simultaneously capture homophily and heterophily graph signals. [65] introduces a universal polynomial basis to enhance the adaptability of spectral GNNs to varying graph structures. While these approaches remain within the spectral paradigm, our work goes one step further by bridging spectral and spatial domains. Instead of limiting our analysis to the spectral view, we establish a coherent mapping from spectral filters to spatial operations, providing deeper insight into the functioning of spectral GNNs. This cross-domain perspective not only advances our GNN understanding but also leads to a naturally derived mechanism that enhances the expressiveness of spectral GNNs beyond prior formulations.

III. NOTATIONS AND PRELIMINARIES

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with node set $\mathcal{V}$ and edge set $\mathcal{E}$, where the number of nodes is denoted by N . The adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ defines the edge weights $A_{i,j}$ between nodes v_i and v_j , and the degree matrix $\mathbf{D}$ can be obtained by summing the rows of $\mathbf{A}$ into a diagonal matrix. We denote the graph Laplacian matrix as $\mathbf{L}$, which is often normalized into $\hat{\mathbf{L}} = \mathbf{I} - \hat{\mathbf{A}}$ with $\hat{\mathbf{A}} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ and an identity matrix $\mathbf{I}$. Let $\hat{\mathbf{L}} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$ be the Laplacian eigendecomposition, where $\mathbf{U}$ is eigenvector matrix and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ consists of eigenvalues within $[0, 2]$. For node classification, we usually have a feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$ with F being raw feature dimensions, and each node is assigned a one-hot label vector $\mathbf{y}_i \in \mathbb{R}^C$ where $C \leq N$ is the class number. Notation details are provided in Nomenclature for easy reading.

A. Spectral Filtering

Spectral filtering is essential in spectral GNNs. It selectively shrinks or amplifies the Fourier coefficients of node features [11] for learning tasks and usually take the form as

$$\mathbf{Z} = g_\psi(\hat{\mathbf{L}}) \mathbf{X} = \mathbf{U} g_\psi(\mathbf{\Lambda}) \mathbf{U}^T \mathbf{X}. \quad (1)$$

Here, $g_\psi : [0, 2] \rightarrow \mathbb{R}$ defines a graph filter function, which is often approximated by a K -order polynomial in practice. Specifically, we have $g_\psi(\lambda) = \sum_{k=0}^K \psi_k P_k(\lambda) = \sum_{k=0}^K \omega_k \lambda^k$ where $P_k : [0, 2] \rightarrow \mathbb{R}$ refers to a polynomial basis and both ψ_k and ω_k denote the polynomial coefficient.

B. Spatial Aggregation

Spatial aggregation is a central component of spatial GNNs, facilitating the propagation of node information along edges and its subsequent aggregation within the node neighborhood.

To provide a better illustration, let's consider a popular spatial GNN, APPNP [44]. This model begins with a feature transformation— $\mathbf{Z}^{(0)} = f(\mathbf{X})$, and then performs the spatial aggregation as

$$\mathbf{Z}^{(k)} = (1 - \eta) \mathbf{Z}^{(0)} + \eta \hat{\mathbf{A}} \mathbf{Z}^{(k-1)}, \quad k = 1, 2, \dots, K \quad (2)$$

where $\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, and η refers to the update rate.

IV. RETHINKING SPECTRAL GNNs FROM THE SPATIAL PERSPECTIVE

In this section, we provide both theoretical and empirical analyses to examine spectral GNNs from the spatial perspective and answer the question, i.e., what information is essentially encoded by spectral GNNs in the spatial domain?

A. Interplay of Spectral and Spatial Domains Through the Lens of Graph Optimization

The graph signal denoising problem [10] was initially leveraged in [25] and [26] as a means to interpret GNNs with a smoothness assumption, which yet does not always hold in certain real-world graph scenarios such as heterophily [53]. Without loss of generality, in this work, we consider a more generalized graph optimization problem¹

$$\arg \min_{\mathbf{Z}} \mathcal{L} = \alpha \|\mathbf{X} - \mathbf{Z}\|_F^2 + (1 - \alpha) \cdot \text{tr}(\mathbf{Z}^T \gamma_\theta(\hat{\mathbf{L}}) \mathbf{Z}) \quad (3)$$

where $\mathbf{Z} \in \mathbb{R}^{N \times d}$ refers to node representations, $\gamma_\theta(\hat{\mathbf{L}})$ determines the rate of propagation [66] by operating on the graph spectrum, i.e., $\gamma_\theta(\hat{\mathbf{L}}) = \mathbf{U} \gamma_\theta(\mathbf{\Lambda}) \mathbf{U}^T$, and $\alpha \in (0, 1)$ is a tradeoff coefficient. In case of setting $\gamma_\theta(\hat{\mathbf{L}}) = \hat{\mathbf{L}}$, (3) turns into the well-known graph signal denoising problem. To ensure the convexity of the objective in (3), a positive semi-definite constraint is imposed on $\gamma_\theta(\hat{\mathbf{L}})$, i.e., $\gamma_\theta(\lambda) \geq 0$ for $\lambda \in [0, 2]$. Then, one can address this minimization problem through either closed-form or iterative solutions.

1) *Closed-Form Solution*: The closed-form solution can be obtained by setting the derivative of the objective function $\mathcal{L}$ to 0, i.e., $\partial \mathcal{L} / \partial \mathbf{Z} = 2\alpha(\mathbf{Z} - \mathbf{X}) + 2(1 - \alpha)\gamma_\theta(\hat{\mathbf{L}})\mathbf{Z} = 0$. Let $g_\psi(\lambda) = (1 + (1 - \alpha)/\alpha\gamma_\theta(\lambda))^{-1}$, we can observe that the closed-form solution in (4) is equivalent to the spectral filtering in (1)

$$\mathbf{Z}^* = \left(\mathbf{I} + \frac{1 - \alpha}{\alpha} \gamma_\theta(\hat{\mathbf{L}}) \right)^{-1} \mathbf{X} = g_\psi(\hat{\mathbf{L}}) \mathbf{X}. \quad (4)$$

As $\gamma_\theta(\lambda) \geq 0$, this establishes a more stringent constraint for the graph filter in spectral GNNs, i.e., $0 < g_\psi(\lambda) \leq \alpha/(\alpha + (1 - \alpha) \cdot 0) = 1$, which is termed as a nonnegative constraint in this article.

2) *Iterative Solution*: Alternatively, we can take an iterative gradient descent method such that $\mathbf{Z}^{(k)} = \mathbf{Z}^{(k-1)} - b \partial \mathcal{L} / \partial \mathbf{Z}|_{\mathbf{Z}=\mathbf{Z}^{(k-1)}}$ with a step size $b = 1/2$, which yields a concise iterative solution in (5) with $\hat{\mathbf{A}}^{\text{new}} = \mathbf{I} - \gamma_\theta(\hat{\mathbf{L}})$. Notably, by taking $\hat{\mathbf{A}}^{\text{new}}$ as a new computation graph, this solution closely mirrors the spatial aggregation in (2)

$$\mathbf{Z}^{(k)} = \alpha \mathbf{X} + (1 - \alpha) \hat{\mathbf{A}}^{\text{new}} \mathbf{Z}^{(k-1)}, \quad k = 1, 2, \dots, K. \quad (5)$$

¹This problem was first introduced in [15] for theoretically grounded graph filters. However, in this study, we repurpose it as a bridge between spectral filtering and spatial aggregation.

3) *Theoretical Interaction—The Adapted New Graph*: With the nonnegative constraint, it is evident that both spectral filtering and spatial aggregation effectively address the generalized graph optimization problem in (3), despite their distinctive forms and operation domains. Upon closer examination, we discover a compelling relationship between the graph filter $g_\psi(\lambda)$ in (4) and the new graph $\hat{\mathbf{A}}^{\text{new}}$ in (5), given $g_\psi(\lambda) = (1 + 1 - \alpha/\alpha\gamma_\theta(\lambda))^{-1}$

$$\hat{\mathbf{A}}^{\text{new}} = \mathbf{I} - \gamma_\theta(\hat{\mathbf{L}}) = \mathbf{I} - \frac{\alpha}{1 - \alpha} \left(g_\psi(\hat{\mathbf{L}})^{-1} - \mathbf{I} \right) \quad (6)$$

which unveils an intrinsic interplay, i.e., spectral filtering implicitly leads the original graph to an adapted new graph, explicitly computed for spatial aggregation.

4) *What Are the Differences Between $\hat{\mathbf{A}}^{\text{new}}$ and $g_\psi(\hat{\mathbf{L}})$* : Whereas the former, as the uncovered new graph elucidates the inherent spatial node relationships, the latter is an operation in spectral GNNs that processes graph features within the spectral domain. It is crucial to understand that $g_\psi(\hat{\mathbf{L}})$ may not result in a dense matrix, especially with fixed-order polynomial approximation. This is because it captures up to only a K -hop neighborhood, i.e., $g_\psi(\hat{\mathbf{L}}) = \sum_{k=0}^K \psi_k P_k(\hat{\mathbf{L}}) = \sum_{k=0}^K \omega_k \hat{\mathbf{A}}^k$, practically limiting spectral GNNs' effective propagation range. In contrast, our newfound graph $\hat{\mathbf{A}}^{\text{new}}$ intrinsically enjoys a nonlocal property, as confirmed in Section IV-B. Building upon this discovery, we further devise a framework to transcend domain barriers, overcoming the limitations of current spectral GNNs due to truncated polynomials (see details in Section V).

B. In-Depth Analysis of the Adapted New Graph

To deepen our understanding of the interpretability produced by spectral GNNs in the spatial domain, we embark upon a blend of theoretical and empirical inquiries into the adapted new graph.

1) *Nonlocality*: Our examination of the adapted new graph illuminates its nonlocal nature, particularly evident in the infinite series expansion of the original graph's adjacency matrix. To elucidate, we first introduce a pivotal mathematical construct, the Neumann series, in the following lemma.

Lemma 1: Let $\mathbf{M} \in \mathbb{R}^{N \times N}$ be a matrix with eigenvalues λ_n , if $|\lambda_n| < 1$ for all $n = 1, 2, \dots, N$, then $(\mathbf{I} - \mathbf{M})^{-1}$ exists and can be expanded as an infinite series, i.e., $(\mathbf{I} - \mathbf{M})^{-1} = \sum_{t=0}^{\infty} \mathbf{M}^t$, which is known as Neumann series.

With the established nonnegative constraint on graph filters, specifically $0 < g_\psi(\lambda) \leq 1$, it becomes evident that the eigenvalues of $\mathbf{I} - g_\psi(\hat{\mathbf{L}})$ falls into the interval permitting Neumann series expansion, as shown in lemma 1 [67]. Building on this observation, we present a nontrivial property of the new graph in the following proposition.

Proposition 1: Given adjacency matrix $\hat{\mathbf{A}}^{\text{new}}$ formulated in (6), the adapted new graph exhibits nonlocality. Specifically, $\hat{\mathbf{A}}^{\text{new}}$ is expressible as an infinite series expansion of the original graph's adjacency matrix $\hat{\mathbf{A}}$. Formally, we have $\hat{\mathbf{A}}^{\text{new}} = \mathbf{I} - \alpha/(1 - \alpha) \sum_{t=1}^{\infty} \left(\mathbf{I} - \sum_{k=0}^K \pi_k \hat{\mathbf{A}}^k \right)^t = \sum_{t=0}^{\infty} \phi_t \hat{\mathbf{A}}^t$ where π_k and ϕ_t refer to the constant coefficients computed from $\{\psi_0, \psi_1, \dots, \psi_K\}$ in distinct ways.

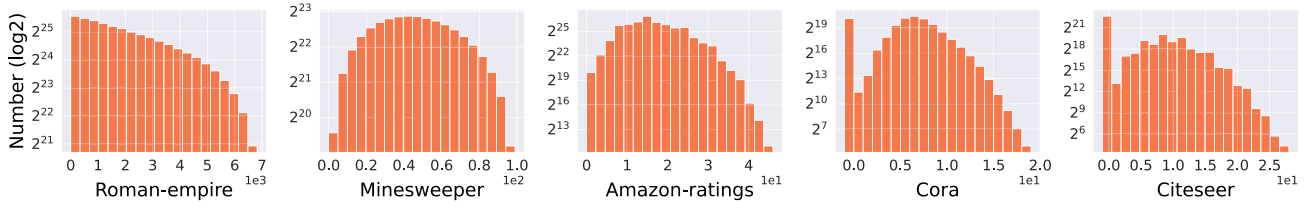


Fig. 1. Distributions of connected nodes in the new graph based on their geodesic/shortest-path distance (as $\Delta_{i,j}$) in the original graph. Nodes, distant in the original graph ($\Delta_{i,j} > 1$ in x-axis), can be linked in the new graph (Number > 0 in y-axis).

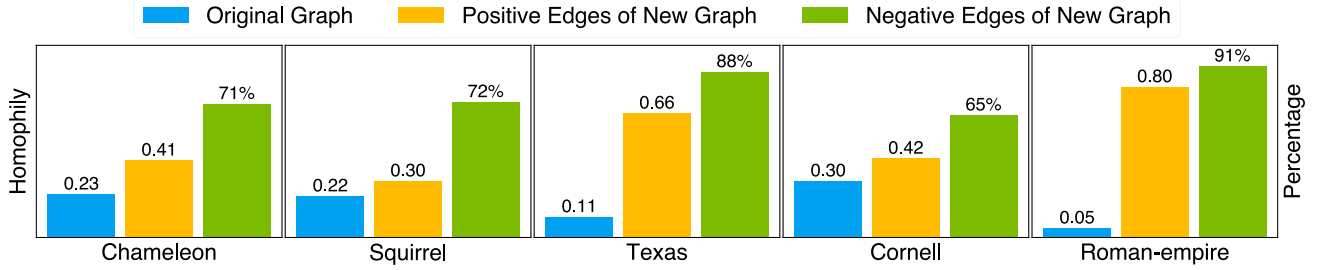


Fig. 2. Left y-axis: homophily comparison between original and new graphs, considering only positive edges (blue and yellow bars). Right y-axis: percentage of negative edges in the new graph that connect nodes from different classes (green bar).

Proof: We begin with the assertion that the eigenvalues of $\mathbf{I} - g_\psi(\hat{\mathbf{L}})$ are positive and strictly less than 1, which fulfills the necessary condition for the Neumann series expansion stated in Lemma 1. As such, we can deduce $g_\psi(\hat{\mathbf{L}})^{-1} = (\mathbf{I} - (\mathbf{I} - g_\psi(\hat{\mathbf{L}})))^{-1} = \sum_{t=0}^{\infty} (\mathbf{I} - g_\psi(\hat{\mathbf{L}}))^t$. Owing to the prevalent polynomial approximation, we are eligible to express $g_\psi(\hat{\mathbf{L}})$ with respect to adjacency matrix $\hat{\mathbf{A}}$, i.e., $g_\psi(\hat{\mathbf{L}}) = g_\psi(\mathbf{I} - \hat{\mathbf{A}}) = \sum_{k=0}^K \pi_k \hat{\mathbf{A}}^k$ where π_k refers to the new coefficients made up of $\{\psi_m\}_{m=0}^K$. Substituting this polynomial representation into our Neumann expansion, we obtain $g_\psi(\hat{\mathbf{L}})^{-1} = \sum_{n=0}^{\infty} (\mathbf{I} - \sum_{k=0}^K \pi_k \hat{\mathbf{A}}^k)^n$. Now, revisiting $\hat{\mathbf{A}}^{\text{new}}$ in (6), we have $\hat{\mathbf{A}}^{\text{new}} = \mathbf{I} - \alpha/(1 - \alpha)(g_\psi(\hat{\mathbf{L}})^{-1} - \mathbf{I}) = \mathbf{I} - \alpha/(1 - \alpha) \sum_{t=1}^{\infty} (\mathbf{I} - \sum_{k=0}^K \pi_k \hat{\mathbf{A}}^k)^t = \sum_{t=1}^{\infty} \phi_t \hat{\mathbf{A}}^t$ where ϕ_t is a constant coefficient made up of $\{\pi_m\}_{m=0}^K$. $\square$

This proposition implies that the new graph engenders immediate links between nodes that originally necessitate multiple hops for connection. To further underpin this theoretical claim, we analyze the general connection status on the new graph by BernNet [15], a spectral GNN adhering to the nonnegative constraint. From Fig. 1, it is apparent that nodes originally separated by multiple hops achieve direct connections in the new graph.

2) *Signed Edge Weights—Discerning Label Consistency:* Upon further scrutinizing the adapted new graph, we make a notable discovery that it readily accommodates both positive and negative edge weights. A more granular analysis in Fig. 2 reveals that a considerable portion of positive edge weights are assigned to the same-class node pairs, enhancing graph homophily (exemplified by edge homophily ratio [53]). Conversely, edges parameterized with negative weights tend to bridge nodes with different labels. These findings demonstrate the newfound graph's adeptness in discerning label consistency among nodes. To theoretically explain this phenomenon, we further present the proposition below.

Proposition 2: Let $\mathbf{Z}^*$ be the node representations optimized by (3). For $\mathbf{Z}^*$ to be effective in label prediction, it is a necessary condition that $\hat{\mathbf{A}}^{\text{new}}$ accommodates both positive and negative edge weights s.t. for any node pairs $v_i, v_j \in \mathcal{V}$, $\hat{A}_{i,j}^{\text{new}} > 0$ if $\mathbf{y}_i = \mathbf{y}_j$ and $\hat{A}_{i,j}^{\text{new}} < 0$ if $\mathbf{y}_i \neq \mathbf{y}_j$.

Proof: Let us commence the proof by contradiction. Let $\mathcal{C}$ denote the condition described in Proposition 2. Assume, for the sake of contradiction, that $\mathcal{C}$ is not requisite for the optimal node representations $\mathbf{Z}^*$ to be predictive of node labels. Under this assumption, there are node pairs $v_i, v_j \in \mathcal{V}$ such that: 1) if $\mathbf{y}_i = \mathbf{y}_j$, $\hat{A}_{i,j}^{\text{new}} < 0$; and 2) if $\mathbf{y}_i \neq \mathbf{y}_j$, $\hat{A}_{i,j}^{\text{new}} > 0$. Without loss of generality, given the nonlocality as proved in Proposition 1, we exclude cases where $\hat{A}_{i,j}^{\text{new}} = 0$ from our consideration. Now, consider the second objective term $\text{tr}(\mathbf{Z}^T \gamma_\theta(\hat{\mathbf{L}}) \mathbf{Z})$ in (3). Using the relationship $\gamma_\theta(\hat{\mathbf{L}}) = \mathbf{I} - \hat{\mathbf{A}}^{\text{new}}$, we can expand this term into $\sum_{v_i, v_j \in \mathcal{V}} \hat{A}_{i,j}^{\text{new}} \|\mathbf{Z}_i - \mathbf{Z}_j\|_2^2$. Under 1), for same-class nodes v_i, v_j with $\hat{A}_{i,j}^{\text{new}} < 0$, minimizing the objective term pulls $\mathbf{Z}_i$ and $\mathbf{Z}_j$ apart in the latent space. This behavior violates the canonical understanding that nodes from the same class should exhibit similar representations. Under 2), for different-class nodes v_i, v_j with $\hat{A}_{i,j}^{\text{new}} > 0$, the optimization encourages $\mathbf{Z}_i$ and $\mathbf{Z}_j$ to be more similar. This is in direct opposition to the basic classification principle that nodes from different classes should have distinct representations. Given these contradictions stemming from the mathematical implications in optimization, we must reject assumptions 1) and 2), affirming the necessary condition $\mathcal{C}$ for accurate label prediction by $\mathbf{Z}^*$. $\square$

Proposition 2 provides a theoretical foundation of our empirical findings on the new graph. The essence lies in the objective in (3), particularly the trace term $\text{tr}(\mathbf{Z}^T \gamma_\theta(\hat{\mathbf{L}}) \mathbf{Z})$. For clarity, let us reinterpret this trace term as $\text{tr}(\mathbf{Z}^T (\mathbf{D}^{\text{new}} - \hat{\mathbf{A}}^{\text{new}}) \mathbf{Z})$, where $\mathbf{D}^{\text{new}}$ denotes the related degree matrix and $\hat{\mathbf{Z}}$ is derived from rescaling $\mathbf{Z}$. Clearly, this term evaluates label smoothness among adjacent nodes in the new graph, which,

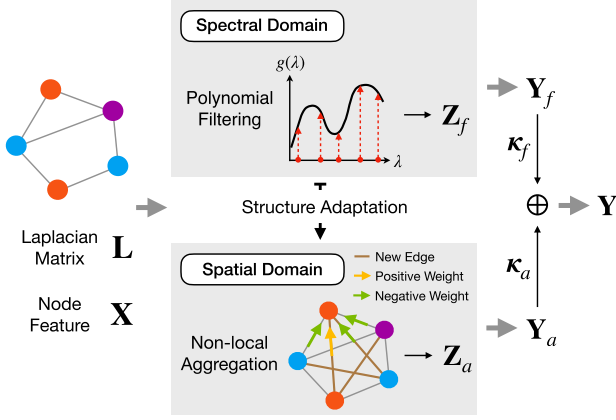


Fig. 3. Illustration of the SAF framework where varying node colors represent different node labels. SAF leverages the adapted new graph by spectral filtering for auxiliary nonlocal aggregation in the spatial domain and allows individual nodes to flexibly balance between spectral and spatial features.

given its nonlocal nature, includes both intra-class (=) and inter-class ($\neq$) node connections such that $\mathbf{A}^{\text{new}} = \mathbf{A}_{=}^{\text{new}} + \mathbf{A}_{\neq}^{\text{new}}$. Drawing from Proposition 2, we can further dissect the original trace term, splitting it into $\text{tr}(\tilde{\mathbf{Z}}^T(\mathbf{D}_{=}^{\text{new}} - \mathbf{A}_{=}^{\text{new}})\tilde{\mathbf{Z}}) - \text{tr}(\tilde{\mathbf{Z}}^T(\mathbf{D}_{\neq}^{\text{new}} - \mathbf{A}_{\neq}^{\text{new}})|\tilde{\mathbf{Z}})$ where the $|\cdot|$ operation denotes absolute values. As such, it becomes evident that minimizing this trace term not only enhances the representational proximity for same-class node pairs but also strengthens the distinctiveness for different-class node pairs. Such nuanced behaviors, inherent to the optimization in (3), are necessary for GNN models to achieve accurate label predictions.

3) *Summarization*: Our investigation into spectral GNNs in the spatial domain reveals that they fundamentally alter the original graph, imbuing it with nonlocality and signed edge weights that capture label relationships among nodes. These findings highlight the interpretable role of spectral GNNs in the spatial domain and prompt us to rethink current models beyond the truncated polynomial filters.

V. SAF FRAMEWORK

Building on our discoveries, we re-evaluate the state-of-the-art spectral GNNs and put forth a paradigm-shifting framework, SAF, for joint exploitation of graph-structured data across both spectral and spatial domains. SAF leverages the adapted new graph by spectral filtering for an auxiliary nonlocal aggregation, addressing enduring challenges in GNNs related to long-range dependencies and graph heterophily. See the overall pipeline in Fig. 3.

A. Nonnegative Spectral Filtering

The proposed SAF requires explicit computation of the newfound graph, as outlined in (6). This further necessitates the graph filter $g_{\psi} : [0, 2] \rightarrow \mathbb{R}$ to satisfy the nonnegative constraint from (3): $0 \leq g_{\psi}(\lambda) \leq 1$. However, not all extant graph filters fulfill this prerequisite. For instance, the filter use by GCN [12], $g_{\psi}(\lambda) = 1 - \lambda$, takes negative values when $\lambda > 1$. In this research, we approximate the graph filter using Bernstein polynomials [68], which are known for

their nonnegative traits [69] and are essential in a preeminent spectral GNN, BernNet [15]. For $g_{\psi}(\lambda) \leq 1$ part, we rescale Bernstein polynomials with the following proposition.

Proposition 3: Let $B_{k,K}(x)$ denote the Bernstein polynomial basis of index k and order K , which is defined as $B_{k,K}(x) = \binom{K}{k}(1-x)^{K-k}x^k$ for $x \in [0, 1]$. Let ψ_k denote the k -th coefficient of a polynomial $p(x)$ of order K , where $p(x) = \sum_{k=0}^K \psi_k B_{k,K}(x)$ with $\psi_k \geq 0$ for all k . Then, for all $x \in [0, 1]$, we have $g_{\psi}(x) \leq \max\{\psi_k\}_{k=0}^K$.

Proof: We denote $p(x) = \sum_{k=0}^K \psi_k \binom{K}{k}(1-x)^{K-k}x^k$ as a Bernstein polynomial with $\psi_k \geq 0$ for all k and $\psi_{\max} = \max\{\psi_k\}_{k=0}^K$. Given $x \in [0, 1]$, we can derive the following inequality as:

$$\begin{aligned} p(x) &= \sum_{k=0}^K \psi_k \binom{K}{k} (1-x)^{K-k} x^k \\ &\leq \psi_{\max} \sum_{k=0}^K \binom{K}{k} (1-x)^{K-k} x^k \\ &= \psi_{\max} (1-x+x)^K = \psi_{\max}. \end{aligned}$$

Therefore, we have $p(x) \leq \max\{\psi_k\}_{k=0}^K$ for all $x \in [0, 1]$. $\square$

Proposition 3 suggests that the Bernstein polynomial function attains its maximum value in $\psi_{\max} = \max\{\psi_k\}_{k=0}^K$. Therefore, $g_{\psi}(\lambda)$ can be rescaled within $[0, 1]$ by $\hat{g}_{\psi}(\lambda) = 1/\psi_{\max} \sum_{k=0}^K \psi_k B_{k,K}(\lambda/2)$, which enables us to formulate the spectral filtering in SAF as $\mathbf{Z}_f = \hat{g}_{\psi}(\hat{\mathbf{L}})f_{\varphi}(\mathbf{X}) = (1/\psi_{\max} \sum_{k=0}^K \psi_k 1/2^K \binom{K}{k} (2\mathbf{I} - \hat{\mathbf{L}})^{K-k} \hat{\mathbf{L}}^k f_{\varphi}(\mathbf{X}))$ where $f_{\varphi}(\cdot)$, a two-layer MLP, maps $\mathbf{X}$ from F to C dimensions using 64 hidden units, and $\{\psi_k\}_{k=1}^K$ are nonnegative learnable parameters. Note that SAF also permits alternative implementations such as using Chebyshev polynomials [11], [70] for graph filter learning, enhancing models like ChebNetII [17] (see details in Section VI-G).

B. Nonlocal Spatial Aggregation

Once acquiring a suitable spectral filter $\hat{g}_{\psi}(\lambda)$, we compute the adapted new graph as $\hat{\mathbf{A}}^{\text{new}} = \mathbf{I} - \tau(\mathbf{U}_m g_{\psi}(\hat{\mathbf{A}}_m)^{-1} \mathbf{U}_m^T - \mathbf{I})$ by (6) where $\tau = \alpha/(1-\alpha)$ is a scaling parameter and a partial eigendecomposition [71] can be employed to produce a low-rank, robust structure for $\hat{\mathbf{A}}^{\text{new}}$ using only m extremal eigenvalues. Equipped with this newfound graph, we proceed to perform nonlocal aggregation: $\mathbf{Z}^{(l)} = (1-\eta)\mathbf{Z}^{(0)} + \eta\hat{\mathbf{A}}^{\text{new}}\mathbf{Z}^{(l-1)}$, where η refers to the update rate and $\mathbf{Z}^{(0)} = f_{\varphi}(\mathbf{X})$ and $l = 1, 2, \dots, L$ denote layer number. The iteratively aggregated results are denoted as $\mathbf{Z}_a$. Recognizing the potential noise from the nonlocal nature of $\hat{\mathbf{A}}^{\text{new}}$, we apply a sparsification technique, leveraging a positive threshold ϵ , and retain only essential elements outside the $[-\epsilon, \epsilon]$ interval. For clarity, this refined model is referred to as SAF- ϵ .

C. Node-Wise Prediction Amalgamation

To leverage information from different graph domains, we employ an attention mechanism, allowing nodes to determine the importance of each space. This mechanism produces pair-wise weights for a nuanced amalgamation during prediction. Specifically, the weight pair is computed

as $\kappa_f = \text{Sigmoid}(\mathcal{P}_f(\mathbf{Z}_f))$, $\kappa_a = \text{Sigmoid}(\mathcal{P}_a(\mathbf{Z}_a))$ where $\kappa_f, \kappa_a \in \mathbb{R}^N$ contain the weights for each node, and $\mathcal{P}_f(\cdot)$ and $\mathcal{P}_a(\cdot)$ are two different mappings from $\mathbb{R}^C$ to $\mathbb{R}$. For simplicity, we implement them using two one-layer MLPs. Given domain predictions $\mathbf{Y}_f, \mathbf{Y}_a \in \mathbb{R}^C$, the final model prediction is attained as $\mathbf{Y} = \text{diag}(\kappa_f) \cdot \mathbf{Y}_f + \text{diag}(\kappa_a) \cdot \mathbf{Y}_a$ where a normalization $[\kappa_f, \kappa_a] \leftarrow ([\kappa_f, \kappa_a]) / (\max(\|[\kappa_f, \kappa_a]\|_1, \delta))$ is performed beforehand to maintain $\kappa_f + \kappa_a = \mathbf{1}$ with small value δ preventing zero division. Similar schemes can be founded in works [72], [73], [74].

D. Complexity Analysis

SAF augments spectral GNNs with nonlocal aggregation and node-wise amalgamation. The first part entails creating a new graph and information propagation. In SAF- ϵ , these two steps are separated, culminating in $\mathcal{O}(N^3 + N^2 + \text{nnz}(\hat{\mathbf{A}}^{\text{new}})d)$ complexity, where nnz denotes nonzero element count. Conversely, SAF, viewing nonlocal aggregation holistically, can reduce complexity to $\mathcal{O}(2dN^2 + dN)$ when $d \ll N$. For node-wise amalgamation, its parallelizable nature ensures computational efficiency. Our method also requires eigendecomposition precomputation, which, though naively complex at $\mathcal{O}(N^3)$, can be reduced to $\mathcal{O}(m^2 + \text{nnz}(\hat{\mathbf{L}})m)$ using the Lanczos method [71] with $m \ll N$ iterative steps. The results are also reusable for both training and inference. We present empirical studies on both time and space overheads in Section VI-J.

E. Practical Implications of Eigendecomposition

Eigendecomposition breaks down a matrix into its eigenvalues and eigenvectors, offering insights into matrix properties, especially for the graph Laplacian. Despite computational demands, this technique has attracted surging interest in the graph learning community due to its theoretical richness, and it can be practically expedited for larger graphs using Lanczos [71] and sparse generalized eigenvalue [75] algorithms. Recent advancements also underscore its value in various applications such as graph positional encoding [76], [77], [78], spectral graph convolution [30], graph domain adaptation [79], graph robustness [80], graph expressivity [29], [58], [81], [82], and graph transformers [50], [83], [84], [85]. In line with these developments, our SAF also utilizes eigendecomposition to explicitly create a new graph, enabling efficient nonlocal aggregation with signed weights to tackle long-range dependency and graph heterophily.

VI. EXPERIMENTS

A. Datasets and Experimental Setup

1) *Datasets*: We conduct experiments on 13 real-world datasets with detailed statistics provided in Table I. In certain compact sections of this article, we use four-letter abbreviations for dataset names.

1) Chameleon and Squirrel are Wikipedia networks collected by [86] where nodes are web pages connected by mutual links. We utilize the labels from [46], dividing nodes into five classes based on their average monthly traffic.

TABLE I

STATISTICS OF REAL-WORLD DATASETS. F AND C DENOTE THE NUMBER OF FEATURES AND CLASSES. Δ REPRESENTS GRAPH DIAMETER REFERRING TO THE LONGEST GEODESIC DISTANCE BETWEEN NODES ON THE GRAPH; FOR PENN94, DUE TO MULTIPLE SUBGRAPHS, WE REPORT Δ OF THE LARGEST CONNECTED COMPONENT. WE ASSESS GRAPH HOMOPIHLY USING THREE METRICS—EDGE HOMOPIHLY [53] $\mathcal{H}$, CLASS HOMOPIHLY [92] $\mathcal{H}_{\text{class}}$, AND ADJUSTED HOMOPIHLY [94] $\mathcal{H}_{\text{adjusted}}$ —WHICH RANGES FROM 0 (HIGH HETEROPIHLY) TO 1 (HIGH HOMOPIHLY)

Dataset	$ \mathcal{V} $	$ \mathcal{E} $	F	C	Δ	$\mathcal{H}$	$\mathcal{H}_{\text{class}}$	$\mathcal{H}_{\text{adjusted}}$
Chameleon	2,227	36,101	2,325	5	11	0.23	0.06	0.03
Squirrel	5,201	217,073	2,089	5	10	0.22	0.03	0.01
Texas	183	309	1,703	5	8	0.11	0.00	-0.23
Cornell	183	295	1,703	5	8	0.30	0.05	-0.08
Actor	7,600	33,544	931	5	12	0.22	0.01	0.00
Cora	2,708	5,429	1,433	7	19	0.81	0.77	0.77
Citeseer	3,327	4,732	3,703	6	28	0.74	0.63	0.67
Pubmed	19,717	44,338	500	3	18	0.80	0.66	0.69
Minesweeper	10,000	39,402	7	2	99	0.68	0.01	0.01
Tolokers	11,758	519,000	10	2	11	0.59	0.18	0.09
Amazon-ratings	24,492	93,050	300	5	46	0.38	0.13	0.14
Roman-empire	22,662	32,927	300	18	6,824	0.05	0.02	-0.05
Penn94	41,554	1,362,229	5	2	8	0.47	0.05	0.02

- 2) Texas and Cornell are webpage datasets from the WebKB project, containing nodes from five classes (student, project, course, staff, and faculty) and connected by hyperlinks. This study uses the preprocessed version by [46].
- 3) Actor is an actor co-occurrence network induced from the film-director-actor-writer network [87] by [46]. In this network, nodes are actors divided into five classes, and edges represent actor co-occurrence on Wikipedia pages.
- 4) Cora, Citeseer, and Pubmed are citation network benchmarks [88], [89] with nodes as scientific papers and edges as undirected citations. Each node is assigned with a class label by research topic and bag-of-words features.
- 5) Minesweeper, Tolokers, Amazon-ratings, and Roman-empire are recently proposed benchmarks by [90] for specifically evaluating GNN models under heterophily.
- 6) Penn94 is a large-scale friendship network [91] from the Facebook 100 networks, where nodes denote students labeled by reported genders and posses features such as major, second major/minor, dorm/house, year, and high school. This work uses the version preprocessed by [92].

2) *Baselines*: We compare SAF with 21 models: 1) MLP; 2) basic GNNs: GCN [12] and APPNP [44]; 3) spectral GNNs: ARMA [41], GPR-GNN [14], BernNet [15], ChebNetII [17], JacobiConv [18], Specformer [50], LON-GNN [20], OptBasis-GNN [21], and FLODE [51]; 4) spatial GNNs: GCNII [27], PDE-GCN [45], NodeFormer [48], GloGNN++ [62], and LRGN [63]; and 5) unified GNNs: GNN-LF [26], GNN-HF [26], ADA-UGNN [25], and FE-GNN [29].

3) *Implementation Details*: To follow [15], [17], and [18], we fix $K = 10$. For each dataset, we perform a grid search to tune the hyperparameters of all models. With the best hyperparameters, we train models with Adam optimizer [93] in 1000 epochs using an early stopping strategy and a patience of 200 epochs, and report the mean classification accuracies

TABLE II
SEMI-SUPERVISED NODE CLASSIFICATION ACCURACY (%)±95% CONFIDENCE INTERVAL

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.
MLP	26.36±2.85	21.42±1.50	32.42±9.91	36.53±7.92	29.75±0.95	57.17±1.34	56.75±1.55	70.52±2.01
GCN	38.15±3.77	31.18±0.93	34.68±9.07	32.36±8.55	22.74±2.37	79.19±1.37	69.71±1.32	78.81±0.84
APPNP	32.73±2.31	24.50±0.89	34.79±10.11	34.85±9.71	29.74±1.04	82.39±0.68	69.79±0.92	79.97±1.58
ARMA	37.42±1.72	24.15±0.93	39.65±8.09	28.90±10.07	27.02±2.31	79.14±1.07	69.35±1.44	78.31±1.33
GPR-GNN	33.03±1.92	24.36±1.52	33.98±11.90	38.95±12.36	28.58±1.01	82.37±0.91	69.22±1.27	79.28±2.25
BernNet	27.32±4.04	22.37±0.98	43.01±7.45	39.42±9.59	29.87±0.78	82.17±0.86	69.44±0.97	79.48±1.47
ChebNetII	43.42±3.54	33.96±1.22	46.58±7.68	42.19±11.61	30.18±0.81	82.42±0.64	69.89±1.21	79.51±1.03
JacobiConv	36.67±1.63	29.38±0.71	48.50±5.90	43.01±11.92	31.69±0.71	82.93±0.55	70.25±1.02	79.53±1.28
Specformer	36.05±3.47	29.64±0.88	50.00±8.33	43.76±5.84	31.45±0.68	81.44±0.63	66.11±0.98	78.05±1.03
LON-GNN	35.17±1.85	30.25±1.04	45.38±7.92	<u>35.32±8.09</u>	31.51±1.23	81.93±0.74	70.41±1.10	79.57±1.08
OptBasisGNN	35.56±2.86	31.25±1.06	37.11±5.09	32.31±7.11	31.73±0.50	78.69±0.86	63.46±1.30	77.38±0.98
FLODE	40.20±1.02	31.99±1.05	<u>50.29±4.74</u>	42.89±7.69	<u>32.18±1.10</u>	79.90±1.07	69.89±2.03	77.78±1.47
GNN-LF	26.49±2.00	22.01±1.04	39.02±6.24	36.65±9.60	28.28±0.71	81.96±0.92	69.80±1.36	79.50±1.28
GNN-HF	35.57±2.26	22.36±1.26	44.80±5.67	38.79±11.62	29.15±0.78	81.15±0.78	69.68±0.73	79.10±1.19
ADA-UGNN	39.39±2.02	25.65±0.49	47.86±6.65	42.89±8.09	30.78±1.00	82.52±1.04	70.18±1.40	79.78±1.32
FE-GNN	38.23±1.66	31.67±1.60	47.40±5.90	41.21±8.96	26.20±0.76	77.00±0.74	61.24±1.26	75.63±1.33
SAF	41.82±1.74	31.77±0.69	58.04±3.76	52.49±8.56	33.50±0.55	83.57±0.66	71.07±1.08	79.51±1.12
SAF- ϵ	41.88±2.04	32.05±0.40	58.38±3.47	53.41±5.55	33.84±0.58	83.79±0.71	71.30±0.93	80.16±1.25
Improv. ³	14.56%	9.68%	15.37%	13.99%	3.97%	1.62%	1.86%	0.68%

with a 95% confidence interval on ten random data splits. As [17] have made a thorough evaluation and share the same experimental protocol with us, we leverage their results for models: MLP, GCN, APPNP, ARMA, GPR-GNN, BernNet, ChebNetII, GCNII, and PDE-GCN on datasets—Chameleon, Squirrel, Texas, Cornell, Actor, Cora, Citeseer, and Pubmed. For JacobiConv, LON-GNN, and OptBasisGNN, we also report the results from their papers [18], [20], [21]. Our codes will be made available if the article could be accepted. For computational requirements, all experiments are conducted on a machine equipped with an NVIDIA GeForce RTX 3090 (24 GB) and an Intel² Xeon² Gold 5218R CPU @ 2.10 GHz (20 cores).

4) *Hyperparameters Setting*: We perform a grid search on the hyperparameters of all models for each dataset using Optuna [95]. To accommodate extensive experiments across diverse datasets in both semi- and full-supervised setting, we define a broad searching space as: learning rate $\text{lr} \sim \{1\text{e-}3, 5\text{e-}3, 1\text{e-}2, 5\text{e-}2, 0.1\}$, weight decay $L_2 \sim \{0.0, 1\text{e-}6, 5\text{e-}6, 1\text{e-}5, 5\text{e-}5, 1\text{e-}4, 5\text{e-}4, 1\text{e-}3, 5\text{e-}3, 1\text{e-}2\}$, dropout $\sim \{0.0, 0.1, \dots, 0.8\}$ with step 0.1, nonlocal aggregation step $L \sim \{1, 2, \dots, 10\}$ with step 1, scaling parameter $\tau \sim \{0.1, 0.2, \dots, 1.0\}$ with step 0.1, update rate $\eta \sim \{0.1, 0.2, \dots, 1.0\}$ with step 0.1, and threshold $\epsilon \sim \{0.0, 1\text{e-}5, 5\text{e-}5, 1\text{e-}4, 5\text{e-}4, 1\text{e-}3, 5\text{e-}3, 1\text{e-}2\}$. For other parameters specific to different base models, we strictly follow their instructions in the original papers.

B. Semi-Supervised Node Classification

In this task, we follow the experimental protocol established by [17] and compare our models with MLP, two basic GNNs, eight popular polynomial spectral GNNs, and five unified GNNs. For data splitting on homophilic graphs (Cora, Citeseer, and Pubmed), we apply the standard division [96] with 20 nodes per class for training, 500 nodes for validation,

and 1000 nodes for testing. On the other five heterophilic graphs, we leverage the sparse splitting [14] with 2.5%/2.5%/95% samples, respectively, for training/validation/testing. The results are reported in Table II, where the best results are bold and the underlined letters denote the second highest accuracy. We first observe that both SAF and SAF- ϵ substantially boosts its base model, BernNet, with gains reaching a notable 15.37%. This impressive enhancement is credited to their capacity to effectively exploit the task-beneficial information, which is implicitly encoded by spectral filtering in the spatial domain. This ability is particularly advantageous in contexts with limited supervision, where it allows effective leveraging of extra prior knowledge during training. Generally, our models outperform competitors on all datasets except for Chameleon and Squirrel, where SAF maintains a second-place rank with considerable improvements on BernNet by 14.56% and 9.68%. The slightly inferior performance compared to ChebNetII on these two datasets may be attributed to their high structural complexity and the relatively limited expressiveness of our base model—BernNet. To better understand SAF’s behavior under such challenging settings, we conduct a further investigation in Section VI-G, where SAF is implemented on top of ChebNetII. This extended analysis provides deeper insights into the expressivity and limitations of our framework and how it interacts with diverse backbones in complex graph scenarios. On the other hand, SAF- ϵ averagely delivers better results than SAF by using thresholding sparsity to reduce nonlocal noise. However, this enhancement also incurs higher computational costs, as illustrated in both Sections V-D and VI-J.

C. Full-Supervised Node Classification

To bolster our evaluation, we expand the previously compared baselines to include five cutting-edge spatial GNN models: GCNII and PDE-GCN capturing long-range dependency, and NodeFormer, GloGNN++ and LRGNN, which

²Registered trademark.

TABLE III
FULL-SUPERVISED NODE CLASSIFICATION ACCURACY (%) $\pm 95\%$ CONFIDENCE INTERVAL

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.
MLP	46.59 $\pm$ 1.84	31.01 $\pm$ 1.18	86.81 $\pm$ 2.24	84.15 $\pm$ 3.05	40.18 $\pm$ 0.55	76.89 $\pm$ 0.97	76.52 $\pm$ 0.89	86.14 $\pm$ 0.25
GCN	60.81 $\pm$ 2.95	45.87 $\pm$ 0.88	76.97 $\pm$ 3.97	65.78 $\pm$ 4.16	33.26 $\pm$ 1.15	87.18 $\pm$ 1.12	79.85 $\pm$ 0.78	86.79 $\pm$ 0.31
APPNP	52.15 $\pm$ 1.79	35.71 $\pm$ 0.78	90.64 $\pm$ 1.70	91.52 $\pm$ 1.81	39.76 $\pm$ 0.49	88.16 $\pm$ 0.74	80.47 $\pm$ 0.73	88.13 $\pm$ 0.33
ARMA	60.21 $\pm$ 1.00	36.27 $\pm$ 0.62	83.97 $\pm$ 3.77	85.62 $\pm$ 2.13	37.67 $\pm$ 0.54	87.13 $\pm$ 0.80	80.04 $\pm$ 0.55	86.93 $\pm$ 0.24
GPR-GNN	67.49 $\pm$ 1.38	50.43 $\pm$ 1.89	92.91 $\pm$ 1.32	91.57 $\pm$ 1.96	39.91 $\pm$ 0.62	88.54 $\pm$ 0.67	80.13 $\pm$ 0.84	88.46 $\pm$ 0.31
BernNet	68.53 $\pm$ 1.68	51.39 $\pm$ 0.92	92.62 $\pm$ 1.37	92.13 $\pm$ 1.64	41.71 $\pm$ 1.12	88.51 $\pm$ 0.92	80.08 $\pm$ 0.75	88.51 $\pm$ 0.39
ChebNetII	71.37 $\pm$ 1.01	57.72 $\pm$ 0.59	93.28 $\pm$ 1.47	92.30 $\pm$ 1.48	41.75 $\pm$ 1.07	88.71 $\pm$ 0.93	80.53 $\pm$ 0.79	88.93 $\pm$ 0.29
JacobiConv	74.20 $\pm$ 1.03	57.38 $\pm$ 1.25	93.44 $\pm$ 2.13	92.95 $\pm$ 2.46	41.17 $\pm$ 0.64	88.98 $\pm$ 0.46	80.78 $\pm$ 0.79	89.62 $\pm$ 0.41
Specformer	75.06 $\pm$ 1.10	65.05$\pm$0.96	90.33 $\pm$ 3.12	90.00 $\pm$ 2.79	42.55 $\pm$ 0.67	88.85 $\pm$ 0.46	80.68 $\pm$ 0.90	91.25 $\pm$ 0.31
LON-GNN	73.00 $\pm$ 2.20	60.61 $\pm$ 1.69	87.54 $\pm$ 3.45	84.47 $\pm$ 3.45	39.10 $\pm$ 1.59	89.44 $\pm$ 1.12	81.41 $\pm$ 1.15	90.98 $\pm$ 0.64
OptBasisGNN	74.26 $\pm$ 0.74	63.62 $\pm$ 0.76	91.15 $\pm$ 1.97	89.84 $\pm$ 2.46	42.39 $\pm$ 0.52	87.96 $\pm$ 0.71	80.58 $\pm$ 0.82	90.30 $\pm$ 0.19
FLODE	74.38 $\pm$ 0.92	63.09 $\pm$ 1.04	89.34 $\pm$ 1.32	89.02 $\pm$ 2.95	<u>42.75$\pm$1.10</u>	88.28 $\pm$ 1.02	80.66 $\pm$ 1.16	90.59 $\pm$ 0.55
GCNII	63.44 $\pm$ 0.85	41.96 $\pm$ 1.02	80.46 $\pm$ 5.91	84.26 $\pm$ 2.13	36.89 $\pm$ 0.95	88.46 $\pm$ 0.82	79.97 $\pm$ 0.65	89.94 $\pm$ 0.31
PDE-GCN	66.01 $\pm$ 1.56	48.73 $\pm$ 1.06	93.24 $\pm$ 2.03	89.73 $\pm$ 1.35	39.76 $\pm$ 0.74	88.62 $\pm$ 1.03	79.98 $\pm$ 0.97	89.92 $\pm$ 0.38
NodeFormer	53.02 $\pm$ 1.58	34.25 $\pm$ 1.96	87.71 $\pm$ 2.13	90.00 $\pm$ 3.45	41.74 $\pm$ 0.61	86.93 $\pm$ 1.22	79.58 $\pm$ 0.85	91.27 $\pm$ 0.39
GloGNN++	72.36 $\pm$ 0.85	60.60 $\pm$ 1.04	91.48 $\pm$ 1.48	89.84 $\pm$ 3.62	41.87 $\pm$ 1.02	87.21 $\pm$ 0.59	79.89 $\pm$ 0.61	86.89 $\pm$ 0.33
LRGNN	75.01 $\pm$ 0.64	63.32 $\pm$ 1.27	91.80 $\pm$ 2.46	90.33 $\pm$ 1.15	41.16 $\pm$ 0.91	88.77 $\pm$ 0.94	79.85 $\pm$ 0.83	90.73 $\pm$ 0.47
GNN-LF	53.74 $\pm$ 1.29	36.15 $\pm$ 0.86	76.07 $\pm$ 2.62	78.36 $\pm$ 2.46	38.39 $\pm$ 0.81	88.51 $\pm$ 0.89	79.84 $\pm$ 0.56	89.86 $\pm$ 0.23
GNN-HF	55.97 $\pm$ 1.05	35.29 $\pm$ 0.72	81.15 $\pm$ 2.62	85.41 $\pm$ 3.12	38.96 $\pm$ 0.77	88.28 $\pm$ 0.64	80.04 $\pm$ 0.93	90.35 $\pm$ 0.30
ADA-UGNN	61.09 $\pm$ 1.51	42.02 $\pm$ 1.26	84.92 $\pm$ 3.12	83.61 $\pm$ 3.44	41.10 $\pm$ 0.62	88.74 $\pm$ 0.85	79.81 $\pm$ 1.11	90.61 $\pm$ 0.44
FE-GNN	73.00 $\pm$ 1.31	63.28 $\pm$ 0.81	88.03 $\pm$ 1.80	86.07 $\pm$ 3.12	41.74 $\pm$ 0.67	89.21 $\pm$ 0.71	80.26 $\pm$ 1.06	90.80 $\pm$ 0.30
SAF	75.30$\pm$0.96	63.63 $\pm$ 0.81	94.10 $\pm$ 1.48	92.95 $\pm$ 1.97	42.93 $\pm$ 0.79	89.80 $\pm$ 0.69	80.61 $\pm$ 0.81	91.49 $\pm$ 0.29
SAF- ϵ	74.84 $\pm$ 0.99	<u>64.00$\pm$0.83</u>	94.75$\pm$1.64	93.28$\pm$1.80	42.98$\pm$0.61	89.87$\pm$0.51	81.45$\pm$0.59	91.52$\pm$0.30
Improv. ³	6.77%	12.61%	2.13%	1.15%	1.27%	1.36%	1.37%	3.01%

go further by not only capturing long-range dependencies but also effectively addressing graph heterophily. For all datasets, we randomly divide them into 60%/20%/20% for training/validation/testing by following [15] and [17]. Table III summarizes the mean classification accuracies. Our methods demonstrate superior performance across most datasets, with an exception on Squirrel, where they only achieve a comparable results to Specformer. This phenomenon can be attributed to the unique characteristics of Squirrel: it contains a substantially larger number of edges (217073), compared to other datasets in Table II (17821 edges on average), making it more complex. In this setting, SpecFormer’s attention mechanism introduces a larger number of parameters, enabling it to better fit the data under the full-supervised regime, where the label rate reaches 60%. However, this benefit comes at a cost—such a design heavily relies on abundant labeled data for effective training. In semi-supervised settings, where only 2.5% of labels are available, this dependence becomes a limitation; as a result, our method outperforms SpecFormer by a notable margin of 2.41% (32.05% versus 29.64%), demonstrating better generalization under a low-label setting.

D. Ablation Study

This section aims to validate our designs by comparing SAF with its three ablated variants—SAF w/o Atte., SAF w/o Spec., SAF w/o Nonloc., and SAF w/o Sign.—in full-supervised node classification. Specifically, Atte., Spec., Nonloc., and Sign., respectively, refer to: attention mechanism in “node-wise prediction amalgamation,” “nonnegative spectral filtering,” and “nonlocal spatial aggregation,” and “signed edge weights.” For SAF w/o Atte., we remove the attention mechanism and

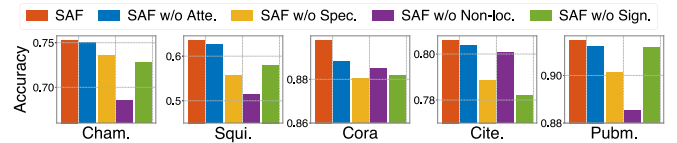


Fig. 4. Ablation study of SAF framework on five datasets.

equally blend predictions from different domains. SAF w/o Spec. abandons the spectral filtering phase, practically setting $\kappa_f = 0, \kappa_a = 1$. As the SAF w/o Spat. configuration is equivalent to the BernNet model, the corresponding results are posted directly. For SAF w/o Sign., we manually prune the negative edges and preserve positive edges only. From Fig. 4, we can draw several insights as follows.

- 1) The impact of Atte. module on our SAF varies by datasets, e.g., on Chameleon and Squirrel, showing a slight performance reduction upon its removal. This observation aligns with our observation that their optimal attention values are close to an even split; conversely, the Cora dataset exhibits a notable drop, due to its optimal attention weights being far from even, as illustrated in Section VI-F.
- 2) Spectral filtering (Spec. module) remains vital for discriminative node representation learning. Specifically, the quality of the adapted new graph fundamentally hinges on the graph spectral filters’ training, as underscored by their theoretical interaction in (6). Practically, the absence of spectral filtering markedly reduces model accuracy, confirming its importance in SAF.

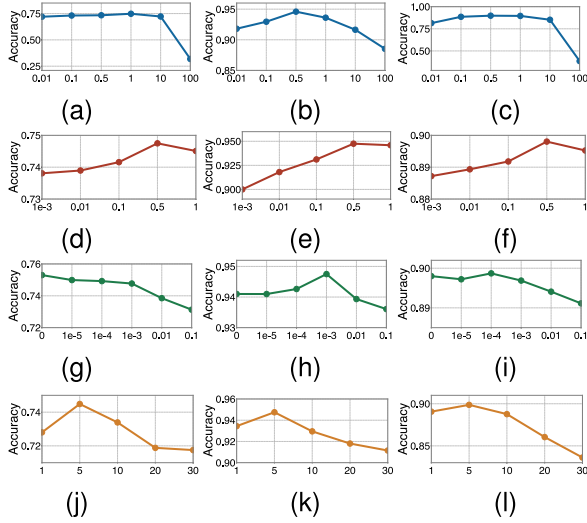


Fig. 5. Sensitivity analysis of hyperparameters: τ , η , ϵ , and L from top to bottom rows. (a) Chameleon. (b) Texas. (c) Cora. (d) Chameleon. (e) Texas. (f) Cora. (g) Chameleon. (h) Texas. (i) Cora. (j) Chameleon. (k) Texas. (l) Cora.

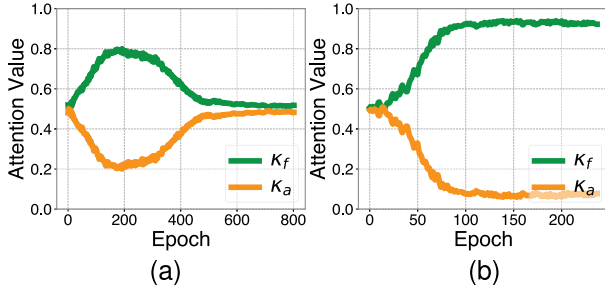


Fig. 6. Attention changing trends with respect to training epochs. (a) Squirrel. (b) Cora.

- 3) This visualization further underscores the importance of the nonlocal aggregation (Nonloc. module), especially on heterophilic and large graphs where capturing long-range dependencies is crucial.
- 4) We highlight the essential role of signed edge weights (Sign. module) in enabling effective nonlocal aggregation. They are critical for capturing label agreement across distant nodes—a key factor in handling heterophily. Notably, when negative edge weights are pruned from the nonlocal graph, we observe a substantial performance degradation, reaffirming their contribution in advancing spectral GNNs.

E. Parameter Study

This section presents the sensitivity analysis of hyperparameters, including τ , η , ϵ , and L . Fig. 5 visualizes how varying these parameters within a broad range influences learning performance, showcasing our model’s robust stability over diverse settings. Beyond empirical observation, we also provide deeper insights into parameter understanding and rationalize the chosen ranges for parameter searching as follows.

- 1) The scaling parameter $\tau = \alpha/(1-\alpha)$, crucial in new graph construction in (6), stems from the tradeoff parameter

$\alpha \in (0, 1)$ within the graph optimization problem in (3). While theoretically we have $0 = 0/(1-0) < \tau < 1/(1-1) = \infty$, practical considerations for extracting structural information suggest a larger penalty on the trace objective term $\text{tr}(\mathbf{Z}^T \gamma_\theta(\hat{\mathbf{L}})\mathbf{Z})$, i.e., keeping $\alpha < 0.5$, thereby limiting $\tau < 0.5/(1-0.5) = 1$. This rationale substantiates our selection of τ within the set $\{0.1, 0.2, \dots, 1\}$ as stated in Section VI-A4, aligning with the observed optimal performance in Fig. 5(a)–(e). When addressing graphs with noisy structure, we may adjust the upper limit of α to $t \in (0, 1)$, setting τ ’s maximum possible value to $t/(1-t)$. For graph benchmarking evaluations in this work, where extracting structural information is important, we practically set $t = 0.5$.

- 2) For the nonlocal aggregation layer number L , a noticeable decline in model performance is observed when L exceeds 10. This is attributed to the nonlocal nature of our new graph, which facilitates efficient information exchange between nodes. Exceeding a certain number of layers may potentially lead to oversmoothing, where there is an overemphasis on global information, thus degrading model performance. However, choosing the number of layers within a reasonable range generally ensures consistent and impressive model performance, as verified in Fig. 5(j)–(l).

F. Analysis of Attention Trends

We analyze the changing trends of the pair-wise attention weights during training SAF on the Squirrel and Cora datasets. From Fig. 6(a) and (b), the average weights for filtering and aggregation start similarly but diverge throughout training, showing different trends in heterophilic and homophilic graphs. On the heterophilic graph Squirrel, both weights converge to similar values, demonstrating their mutual importance in modeling complex connectivity. Conversely, κ_f becomes dominant on the homophilic graph Cora due to the sufficiency of node proximity information for label prediction, thereby diminishing the relevance of κ_a and nonlocal aggregation.

G. SAF With ChebNetII as Base Model

To show the versatility of our SAF framework, we introduced ChebNetII [17] as an alternative base model, chosen for its adherence to the nonnegative constraint, critical in our model design as stated in Section V. The rationale behind this choice is ChebNetII’s use of Chebyshev interpolation for learning Chebyshev polynomials, where the constraint can be ensured by keeping its learnable parameters $\{\gamma_j\}_{j=0}^K$ nonnegative. In a full-supervised setting (see Table IV), SAF consistently improves ChebNetII, validating the adaptability of our framework across different spectral filters. However, in a semi-supervised setting (see Table V), the improvements of SAF-Cheb over ChebNetII are less significant compared to the gains achieved by SAF-Bern over BernNet, and even slightly underwhelming in certain cases, such as on the Squirrel dataset. This phenomenon could be attributed to the $g_\phi(\lambda) \leq 1$ constraint within SAF (refer to Section V), necessitating the rescaling of filter functions by their maximum values.

TABLE IV
FULL-SUPERVISED NODE CLASSIFICATION ACCURACY (%) WHILE IMPLEMENTING SAF UPON ChebNetII

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.
BernNet	68.53±1.68	51.39±0.92	92.62±1.37	92.13±1.64	41.71±1.12	88.51±0.92	80.08±0.75	88.51±0.39
SAF-Bern	75.30±0.96	63.63±0.81	94.10±1.48	92.95±1.97	42.93±0.79	89.80±0.69	80.61±0.81	91.49±0.29
SAF-Bern- ϵ	74.84±0.99	64.00±0.83	94.75±1.64	93.28±1.80	42.98±0.61	89.87±0.51	81.45±0.59	91.52±0.30
ChebNetII	71.37±1.01	57.72±0.59	93.28±1.47	92.30±1.48	41.75±1.07	88.71±0.93	80.53±0.79	88.93±0.29
SAF-Cheb	74.97±0.66	64.06±0.59	94.43±1.81	92.62±2.13	42.65±1.01	89.56±0.64	80.68±0.68	91.27±0.34
SAF-Cheb- ϵ	75.25±0.96	64.42±0.82	94.26±1.64	93.12±1.64	42.79±1.04	89.61±0.71	81.08±0.68	91.73±0.18

TABLE V
SEMI-SUPERVISED NODE CLASSIFICATION ACCURACY (%) WHILE IMPLEMENTING SAF UPON ChebNetII

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.
BernNet	27.32±4.04	22.37±0.98	43.01±7.45	39.42±9.59	29.87±0.78	82.17±0.86	69.44±0.97	79.48±1.47
SAF-Bern	41.82±1.74	31.77±0.69	58.04±3.76	52.49±8.56	33.50±0.55	83.57±0.66	71.07±1.08	79.51±1.12
SAF-Bern- ϵ	41.88±2.04	32.05±0.40	58.38±3.47	53.41±5.55	33.84±0.58	83.79±0.71	71.30±0.93	80.16±1.25
ChebNetII	43.42±3.54	33.96±1.22	46.58±7.68	42.19±11.61	30.18±0.81	82.42±0.64	69.89±1.21	79.51±1.03
SAF-Cheb	43.93±1.54	32.52±0.68	56.36±2.83	48.90±4.68	33.54±0.51	83.05±0.91	70.94±0.95	79.83±1.18
SAF-Cheb- ϵ	44.17±1.52	32.47±0.45	56.99±4.05	47.23±7.17	33.66±0.23	83.45±0.62	71.27±0.90	80.00±1.11

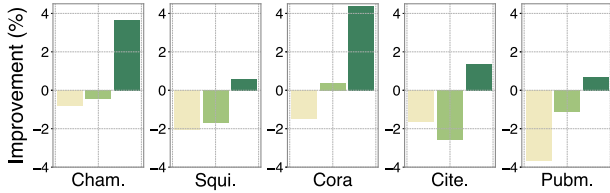


Fig. 7. Improved classification accuracy by GloGNN++, LRGNN, and our SAF (from left to right) over the vanilla GCN under semi-supervised setting (2.5%/2.5%/95%).

For Bernstein polynomials, this maximum is readily obtained as the largest polynomial coefficient $\max\{\phi_k\}_{k=0}^K$, as per Proposition 3. However, for Chebyshev polynomials, the best theoretical upper bound is the sum of absolute coefficients, $\sum_{k=0}^K |\phi_k|$, which is comparatively less precise—particularly in semi-supervised settings, where filter parameters may not be sufficiently and accurately optimized due to limited labeled data. This difference may impact the quality of graph construction and, subsequently, compromise the model’s performance. Exploring these nuances will be a focal point of our future research.

H. Supervision Dependence in GloGNN++ and LRGNN

This section empirically verifies the heavy reliance on label supervision for both GloGNN++ [62] and LRGNN [63] models. In Fig. 7, we compare the improved classification accuracy of these models and our SAF upon the vanilla GCN under a semi-supervised setting. The figure illustrates that, despite competitive results under dense supervision (see Table III), the performance of both GloGNN++ and LRGNN deteriorates, demonstrating negative optimization on many datasets, when the label rate drops approximately from 60% to 2.5%. This degradation highlights their dependency on high label rates for effective learning. In contrast, our SAF

constantly delivers good performance regardless of the supervision level, showing its effectiveness in both high and low-label rate scenarios.

I. New Benchmarks for Graph Heterophily

For a more extensive evaluation across various domains, we also test SAF on five recently introduced datasets, including Mine., Tolo., Amaz., Roma., and Penn94 [90], [92]. In this context, we draw comparisons solely with MLP, GCN, APPNP, along with seven GNN models that have previously shown promising results in prior tasks, namely, GPR-GNN, BernNet, ChebNetII, JacobiConv, NodeFormer, GloGNN++, and FE-GNN. Table IX lists the average classification accuracies, obtained over random splits provided by [90] and [92], with a distribution of 50%/25%/25% for training/validation/testing. In summary, SAF achieves significant performance gains of 12.79% and 4.03% on Minesweeper and Tolokers datasets, respectively, while maintaining competitiveness on the others.

J. Time and Space Overheads

1) *Eigendecomposition*: Our SAF framework precomputes eigendecomposition once per graph and reuses it in (6) in both training and inference. This aspect is crucial, as the forward-pass cost in model training often exceeds the preprocessing expense of eigendecomposition. To empirically validate this, we compare the time overheads of eigendecomposition with the training times of various models in Table VI. Overall, we have following observations: 1) for datasets with a small number of nodes, the time consumed by decomposition is significantly less than the time required for model training; 2) for medium-sized graphs such as Pubmed, while the full decomposition time exceeds that of BernNet, it still maintains efficiency against more advanced GNNs such as ChebNetII, NodeFormer and GloGNN++; and 3) moving to the large-scale graph, Penn94 (with 41 554 nodes and 1 362 229 edges),

TABLE VI
TIME OVERHEADS (s)

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.	Penn94
# Nodes	2,277	5,201	183	183	7,600	2,708	3,327	19,717	41,554
# Edges	36,101	217,073	309	295	33,544	5,429	4,732	44,338	1,362,229
BernNet	8.36	13.74	3.92	4.16	4.88	5.24	5.52	6.06	24.05
ChebNetII	22.82	30.73	11.47	9.64	14.88	19.96	16.14	36.91	41.67
FE-GNN	3.99	45.89	1.03	0.84	0.98	2.54	2.06	6.78	7.33
NodeFormer	58.96	79.66	14.29	18.89	66.20	19.25	32.00	68.57	122.91
GloGNN++	35.63	68.31	4.47	3.00	73.13	32.68	12.35	5266.53	3614.37
SAF	11.55	18.78	4.38	4.70	5.36	6.04	6.12	18.43	23.49
Decomposition	0.58	1.59	0.02	0.02	3.93	1.00	0.77	21.34	4.76

TABLE VII
SPACE OVERHEADS (MB)

Method	Cham.	Squi.	Texas	Corn.	Actor	Cora	Cite.	Pubm.	Penn94
BernNet	72	232	5	5	292	64	152	1546	1902
ChebNetII	72	231	5	5	291	63	152	1584	1850
FE-GNN	1337	6919	23	10	416	302	740	2213	5854
NodeFormer	1522	3965	15	37	775	480	764	2119	3056
GloGNN++	290	1525	5	5	2471	331	607	17892	25260
SAF	112	440	5	5	733	120	237	4515	8491
Decomposition	141	540	1	1	1206	140	239	7641	4

TABLE VIII

ANALYSIS OF SAF USING PARTIAL DECOMPOSITION WITH PUBMED DATASET AS AN EXAMPLE. BASELINE AVERAGES ARE COMPUTED FOR OVERALL COMPARISON. FOR FAIR COMPARISON, TRIVIAL MODELS (MLP, GCN, AND APPNP) ARE EXCLUDED FOR ACCURACY, AND GloGNN++ IS EXCLUDED FOR TIME OVERHEADS DUE TO EXTREME RESOURCE USAGE

Method	Accuracy (%)	Model Time (S)	Decomposition Time (S)	Decomposition Space (MB)
Baselines w/o {MLP, GCN, APPNP}	89.77	-	-	-
Baselines in Table VII w/o {GloGNN++}	89.88	29.58	-	-
SAF (using 19,717 eigenpairs)	91.49±0.29	18.43	21.34	7641
SAF (using 2,000 eigenpairs)	90.80±0.41	15.45	16.34	362
SAF (using 1,600 eigenpairs)	90.18±0.38	14.51	15.30	358
SAF (using 1,200 eigenpairs)	90.10±0.04	12.99	12.31	355
SAF (using 800 eigenpairs)	90.05±0.04	12.02	9.95	358

TABLE IX
EVALUATIONS ON NEW HETEROPHILIC GRAPH DATASETS

Method	Mine.	Tolo.	Amaz.	Roma.	Penn94
MLP	50.61±0.87	74.58±0.69	45.50±0.38	66.11±0.33	74.58±0.37
GCN	72.25±0.60	76.56±0.85	48.06±0.39	53.49±0.33	82.47±0.27
APPNP	68.48±1.20	74.13±0.62	48.12±0.37	72.99±0.46	75.29±0.27
GPR-GNN	89.76±0.53	75.82±0.50	49.06±0.25	73.19±0.24	81.38±0.16
ChebNetII	83.62±1.51	78.95±0.49	49.76±0.36	74.52±0.54	83.12±0.22
JacobiConv	89.88±0.33	77.24±0.39	43.89±0.28	74.30±0.50	83.35±0.11
NodeFormer	89.89±0.46	80.31±0.75	43.67±1.54	73.59±0.60	69.66±0.83
GloGNN++	72.59±1.54	79.01±0.84	50.03±0.29	66.10±0.26	73.15±0.59
FE-GNN	84.68±0.36	79.31±0.59	49.46±0.29	74.50±0.30	82.30±0.54
BernNet	77.75±0.61	75.35±0.63	49.84±0.52	<u>74.56±0.74</u>	82.47±0.21
SAF	90.54±0.30	<u>79.38±0.58</u>	50.49±0.28	74.87±0.22	83.86±0.26
Improv.	12.79%	4.03%	0.65%	0.31%	1.39%

where only partial eigendecomposition with 100 extremal eigenvalues is considered, the computation time is markedly reduced compared to all the models. In terms of space overheads, we have the following observations: 1) when focusing

on the eigendecomposition itself across different datasets, the trend in space overheads generally aligns with that of time overheads—both increase with dataset scale. This is because eigendecomposition requires storing intermediate matrixes whose sizes grow with node and edge number and 2) when comparing eigendecomposition to models, the space overhead patterns are not always consistent with the time overheads. For instance, on the complex datasets such as Squirrel, although eigendecomposition achieves lower time overhead (due to its one-time computation), its peak memory usage still surpasses most models. This is because the space overhead is determined by the eigendecomposition step itself and cannot be amortized over reuse, unlike the time cost. To mitigate this limitation, our SAF framework supports partial eigendecomposition, which computes only a small subset of extremal eigenpairs, thereby significantly reducing computational demands. As shown in Table VIII, we empirically validate this on the Pubmed dataset (19 717 nodes) as an example. By using only 2000 eigenvalues, the space overhead drops from 7641 to 362 MB—markedly

lower than all baselines reported in Table VII—with only a 0.69% drop in accuracy. Moreover, as the number of eigenpairs decreases, we observe a notable reduction in decomposition time, dropping by 53% (from 21.34 to 9.95 s), which is also significantly below the average baseline running time (29.58 s) reported in Table VI. In addition, SAF's overall model runtime also decreases accordingly, since using partial decomposition reduces the complexity from $\mathcal{O}(2dN^2 + dN)$ to $\mathcal{O}(2dNm + dm)$, where m denotes eigenpair number and $m \ll N$. Despite these simplifications, our model maintains competitive accuracy and consistently outperforms the average baseline performance.

2) *Model Comparison*: In Table VI, we compare the running times of our SAF against two notable spectral GNNs (BernNet and ChebNetII), two nonlocal GNNs (NodeFormer and GloGNN++), and one unified GNN model (FE-GNN). Generally, one can observe that SAF, while slightly slower than its base model, BernNet, due to the integration of nonlocal spatial aggregation, remains more efficient than or comparable to other SOTA methods, particularly those also employing nonlocal approaches such as NodeFormer and GloGNN++.

VII. CONCLUSION

This study presents a cross-domain analysis of GNN models, offering a fresh perspective by rethinking spectral GNNs from a spatial lens. We reveal that spectral GNNs fundamentally lead the original graph to an adapted new one, which exhibits nonlocality and accommodates signed edge weights to reflect label consistency among nodes. This insight leads to our proposed SAF framework, enhancing spectral GNNs for more effective and versatile graph representation learning. While SAF adeptly captures long-range dependencies and addresses graph heterophily, we acknowledge two limitations of this work: 1) the nonnegative constraints of the proposed SAF on graph filters might limit model expressiveness, indicating room for theoretical refinement; and 2) although this study focuses on a node-level investigation, it raises intriguing questions about the implications of spectral GNNs at the graph-level in the spatial domain. Future work could expand this examination by relaxing theoretical constraints or exploring the cross-domain interplay from a broader graph-level viewpoint.

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